

LABORATORY OBSERVATION /RECORD

Course: Full stack web development

Course code: 23CM4121

Experiment No: 01

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1. TITLE

Develop a College Information Portal Webpage Using HTML5

2. AIM

To develop a mini web application using HTML that accepts student information through an HTML form and displays the submitted details.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure and elements of HTML5.
 - To create a well-structured college information webpage using HTML.
 - To apply text formatting, lists, hyperlinks, tables, and images.
 - To design a student registration form using HTML form elements.
 - To use HTML5 semantic elements for organizing webpage content.
 - To implement multimedia elements such as audio and video.
 - To use modern HTML5 input types such as email, date, number, color, and range.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages.

HTML5 is the latest version of HTML that provides semantic elements, multimedia support, forms, and modern input types.

In this application:

- HTML5 document structure is used to organize the webpage.
- Headings and paragraphs are used to display college information.
- Lists are used to display courses and facilities.
- Tables are used to display student details.
- Images are inserted using the `<img>` tag.
- Forms are used to collect student registration details.
- Semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` provide meaningful page structure.
- Audio and video are embedded using `<audio>` and `<video>`.
- HTML5 input types such as email, date, number, color, and range provide specialized form controls.

The basic structure of the webpage is:

HTML5 Document → Header → Navigation → Content Sections → Forms → Multimedia → Footer

5. ALGORITHM / PROCEDURE

1. Create a valid HTML5 document using `<!DOCTYPE html>`.
2. Add the page title and college information using headings and paragraphs.
3. Apply text formatting and create ordered and unordered lists.
4. Add hyperlinks to the college website and Google.
5. Insert the college logo and campus image with appropriate alternate text.
6. Create a student details table with Roll No, Name, Branch, and Email.
7. Design a student registration form with text, email, password, radio buttons, checkboxes, dropdown, textarea, and submit/reset controls.
8. Add semantic HTML5 elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>`.
9. Add HTML5 input types such as date, number, color, and range.
10. Embed audio and video using the `<audio>` and `<video>` elements.
11. Save the file with an .html extension and open it in a web browser.
12. Verify that all links, images, form controls, and multimedia elements work correctly.

6. Program

Part1.html

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>HTML5 Canvas</title>

</head>

<body>


  <h1>HTML5 Canvas Example</h1>


  <canvas id="myCanvas" width="500" height="300" style="border:1px solid black;">

    Your browser does not support the canvas element.

  </canvas>


  <script>

    var canvas = document.getElementById("myCanvas");

    var ctx = canvas.getContext("2d");


    // Rectangle

    ctx.fillStyle = "lightblue";

    ctx.fillRect(20, 20, 120, 70);


    // Circle

    ctx.beginPath();

    ctx.arc(220, 55, 35, 0, 2 * Math.PI);

    ctx.stroke();
```

```
// Line
```

```
ctx.beginPath();
```

```
ctx.moveTo(20, 150);
```

```
ctx.lineTo(250, 150);
```

```
ctx.stroke();
```

```
// Text
```

```
ctx.font = "24px Arial";
```

```
ctx.fillStyle = "blue";
```

```
ctx.fillText("suswetha", 20, 230);
```

```
</script>
```

```
</body>
```

```
</html>
```

```
part2.html
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>CSS3 Styling</title>
```

```
<style>
```

```
body{
```

```
margin:0;
```

```
font-family:Arial, sans-serif;

background:#f4f4f4;

}
```

```
/* Header */
```

```
header{

    background:linear-gradient(to right,#4facfe,#00f2fe);

    color:white;

    text-align:center;

    padding:20px;

    text-shadow:2px 2px 4px gray;

    font-size:30px;

}
```

```
/* Navigation */
```

```
nav{

    background:#333;

    text-align:center;

    padding:12px;

}
```

```
nav a{

    color:white;

    text-decoration:none;

    margin:20px;
```

```
    font-size:18px;
}
```

```
nav a:hover{
    color:yellow;
}
```

```
/* Cards */
```

```
.container{
    display:flex;
    justify-content:space-around;
    margin:20px;
}
```

```
.card{
    width:28%;
    background:white;
    border:2px solid #ddd;
    border-radius:10px;
    padding:20px;
    box-shadow:5px 5px 10px gray;
    text-align:center;
}
```

```
.card h2{
```

```
font-family:Georgia, serif;  
}
```

```
.card button{  
    background:#007BFF;  
    color:white;  
    border:none;  
    padding:10px 20px;  
    border-radius:5px;  
    cursor:pointer;  
}
```

```
.card button:hover{  
    background:green;  
}
```

```
/* Footer */  
footer{  
    background:#222;  
    color:white;  
    text-align:center;  
    padding:15px;  
    margin-top:20px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
    CSS3 Webpage Design
```

```
</header>
```

```
<nav>
```

```
    <a href="#">Home</a>
```

```
    <a href="#">About</a>
```

```
    <a href="#">Services</a>
```

```
    <a href="#">Contact</a>
```

```
</nav>
```

```
<div class="container">
```

```
    <div class="card">
```

```
        <h2>Card 1</h2>
```

```
        <p>This is the first content card.</p>
```

```
        <button>Read More</button>
```

```
    </div>
```

```
    <div class="card">
```

```
        <h2>Card 2</h2>
```

```
<p>This is the second content card.</p>

<button>Read More</button>

</div>

<div class="card">

  <h2>Card 3</h2>

  <p>This is the third content card.</p>

  <button>Read More</button>

</div>

</div>

<footer>

  © 2026 CSS3 Styling Example

</footer>

</body>

</html>
```

7 . CODE EXPLANATION

Code	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.

<code><html lang="en"></code>	Defines the root element and specifies English as the document language.
<code><head></code>	Contains metadata, the page title, and the CSS stylesheet reference.
<code><meta charset="UTF-8"></code>	Specifies UTF-8 character encoding.
<code><meta name="viewport"></code>	Helps the webpage adapt to different screen sizes.
<code><title></code>	Sets the title displayed in the browser tab.
<code><link rel="stylesheet" href="style.css"></code>	Connects the external CSS file to the HTML document.
<code><div class="header"></code>	Creates the header section of the webpage.
<code><div class="navbar"></code>	Creates the navigation bar containing navigation links.
<code></code>	Creates a hyperlink.
<code><div class="hero"></code>	Creates the main introductory section of the webpage.
<code><div class="canvas-section"></code>	Creates the section containing the HTML5 Canvas.
<code><canvas id="myCanvas" width="700" height="250"></code>	Creates a 700 × 250 pixel drawing area for Canvas graphics.
<code><div class="cards-container"></code>	Contains the three content cards.

<code><div class="card"></code>	Defines an individual content card.
---------------------------------------	-------------------------------------

<code><button></code>	Creates an interactive button for each card.
<code><div class="footer"></code>	Creates the footer section of the webpage.
<code><script></code>	Contains the JavaScript code used to draw graphics on the Canvas.
<code>document.getElementById("myCanvas")</code>	Retrieves the Canvas element using its ID.
<code>canvas.getContext("2d")</code>	Obtains the 2D drawing context used to draw on the Canvas.
<code>ctx.fillStyle</code>	Sets the color used for filling shapes or text.
<code>ctx.fillRect()</code>	Draws a filled rectangle on the Canvas.
<code>ctx.beginPath()</code>	Starts a new drawing path.
<code>ctx.arc()</code>	Creates a circular arc, which is used to draw the circle.
<code>ctx.fill()</code>	Fills the current path with the selected color.
<code>ctx.moveTo()</code>	Moves the drawing cursor to a specified position without drawing.
<code>ctx.lineTo()</code>	Creates a line from the current position to the specified position.
<code>ctx.closePath()</code>	Closes the current drawing path.
<code>ctx.stroke()</code>	Draws the outline of the current path.

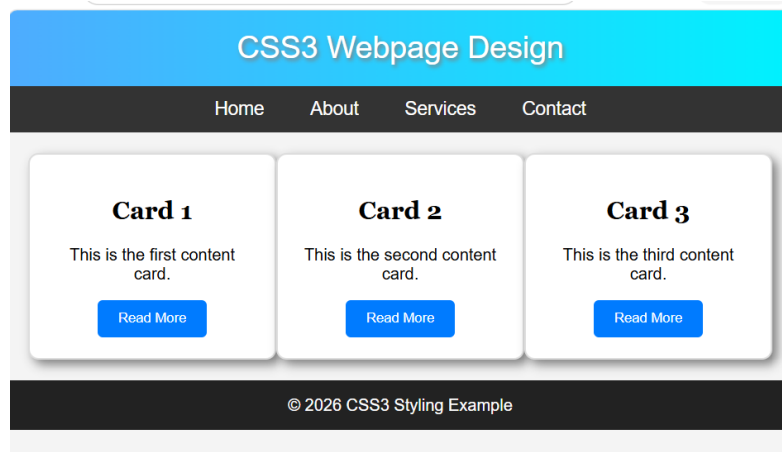
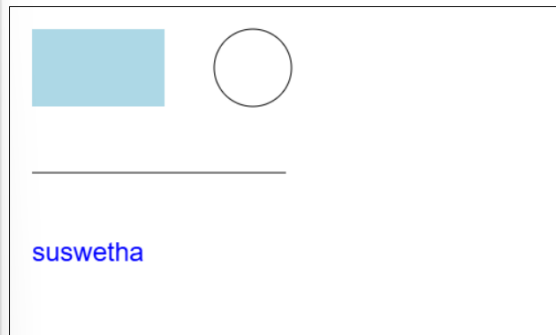
ctx.strokeStyle	Sets the color of the line.
ctx.lineWidth	Sets the thickness of the line.
ctx.font	Specifies the font style and size used for Canvas text.
ctx.fillText()	Displays text at a specified position on the Canvas.

8. TEST CASES AND EXECUTION DATA

Test Case	Expected Result	Actual Result	Status	Test Case	Expected Result	Actual Result
Open the HTML file in a browser	Webpage loads with header, navigation, cards, and footer	Webpage loaded correctly	Pass	Open the HTML file in a browser	Webpage loads with header, navigation, cards, and footer	Webpage loaded correctly
Check Canvas graphics	Rectangle, circle, triangle, line, and name are displayed	All graphics displayed correctly	Pass	Check Canvas graphics	Rectangle, circle, triangle, line, and name are displayed	All graphics displayed correctly
Hover over navigation links	Link color and border change smoothly	Hover effect works correctly	Pass	Hover over navigation links	Link color and border change smoothly	Hover effect works correctly

9. OUTPUT

HTML5 Canvas Example



10. RESULT

Thus, a styled webpage was successfully developed using HTML5 Canvas and CSS3. Basic graphics, responsive layout, CSS3 styling properties, and interactive hover effects were implemented successfully, and the webpage was verified using multiple test cases.

10. RESULT